

Supervised Non-negative Matrix Factorization for Audio Source Separation

Abstract

This paper introduces a supervised non-negative Matrix Factorization pipeline for music source separation, refining the Kullback-Leibler divergence (KL) cost function to isolate individual stems (drums, vocals, bass, etc.). Current NMF architectures treat spectrogram frames as independent observations, often leading to a loss of temporal structure and inconsistent results. To address the main contingencies with the traditional approach, we propose a two-stage pipeline that leverages targeted matrix constraints, decoupling regularization, and preprocessing techniques to yield more interpretable results.

In the supervised dictionary training phase, we extract non-silent frames from the MUSDB18 dataset and process the basis through a high-pass filter. During the separation phase, we fix the global dictionary, and the weight matrix is optimized using modified multiplicative update rules. An L_1 sparsity penalty is introduced to transient sources to suppress noise floors, and a variance-normalized Temporal Continuity (TC) penalty is applied to harmonic sources to promote smooth, sustained series. Finally, source reconstruction is achieved by applying Wiener Filtering to the original complex Short-Time Fourier Transform (STFT) to recover phase information. Quantitative evaluation metrics such as **SIR**, **SAR** and **SDR** on the reconstructed stems demonstrated tailored rank allocation and decoupled regularization correspond to higher-fidelity sound separation.

1 Introduction

In real world scenarios, audio signals are typically composed of concurrent sound sources. Consequently, audio source separation has emerged as a critical enabling technology for numerous downstream applications. Examples range from isolating speakers for hearing aids, enhancing voice commands with devices, and transcribing polyphonic music. In this paper, we will focus specifically on the task of music source separation.

Foundational NMF and Audio Representation

Traditional NMF has been shown to be explainable for extracting features from high-dimensional datasets. Its non-negativity constraint generates interpretable matrices in applications such as facial recognition, signal processing, and text mining; actions that inherently abide by the constraint. Lee and Seung's foundational work on NMF and Multiplicative Update Rules (MU) translates gradient descent to safely factorize strictly positive data without boundary violations. In audio signal processing, we convert time-domain audio to magnitude spectrograms via the Short-Time Fourier Transform, satisfying NMF's non-negativity constraint. However, established findings have suggested switching from Euclidean distance to

KL divergence, highlighting KL’s scale-invariant properties, which better model low-energy observations, unlike Euclidean distance, which assumes Gaussian Noise [5]. As a result, KL divergence makes better estimations on audio perceptions.

Weakness of regular NMF

Although KL-NMF yields a better solution to the source separation problem, it treats each time frame of the spectrogram as an independent observation. Conceptually, this ignores the intrinsic, highly correlated, temporal nature of music, resulting in ”unstable” activation weights. Additionally, because KL-NMF relies on magnitude spectrograms, the reconstruction sounds almost ”robotic” due to the discarded phase information.

Prior Attempts at Regularization

Present-day solutions to the independence problem include adding a temporal continuity cost that smooths harmonic activations over time. Moreover, for transient instruments such as drums, minimizing the L_1 norm promotes sparsity in the W matrix. Current approaches apply these measures at the cost of **uniform application**. Applying the same regularization restraints across all instruments results in ”penalty collapse”, where high-variance signals such as vocals get collapsed under the penalties, resulting in energy leakage to unpenalized portions of the basis matrix, which degrades separation quality.

Our Approach

The pipeline described in this paper attempts to solve these contingencies. Rather than a constant regularization cost across all instruments, we implement instrument-specific, decoupled regularization tailored to specific acoustic properties. Furthermore, during dictionary training, we use high-pass filtering to enforce orthogonality in our matrix, removing low-frequency templates from the vocal/harmonic matrices and addressing the basis coherence problem. In our final reconstruction, we use the NMF estimate to create a Wiener filter applied to the original complex STFT to re-inject the original phase, ensuring energy conservation and phase recovery.

2 Related Work

2.1 Foundational Non-negative Matrix Factorization

Lee and Seung’s original paper introduces NMF and the multiplicative update rules, establishing the foundational iterative optimization scheme [2]. Using such algorithms has shown success in parts-based representation such as facial recognition [4]. Subsequent research by Smaragdis and Virtanen pioneered the application of NMF to audio by treating Short-Time-Fourier Transform (STFT) magnitude spectrograms as non-negative datasets [9].

2.2 Cost Functions and Acoustic Perception

Early models relied on the Euclidean norm, which assumes Gaussian noise and penalizes high energy errors more heavily [3]. As a result, the KL divergence minimization problem was subsequently shown to be superior for audio [5]. The scale invariant property maps better to the logarithmic perception of human hearing, preserving quiet harmonics alongside loud transients.

2.3 Regularization Constraints in Source Separation

Virtanen’s paper and similar works identified standard NMF’s failure to account for time. Standard NMF treats each frame independently, causing jittery and less interpretable activations. To address this, Temporal Continuity (TC) penalties were introduced to enforce smoothness in harmonic instruments (vocals, bass, etc.) [8]. Sparsity constraints have been widely adopted to improve the found decompositions. [6]. However, these regularizations are applied globally, where uniform application forces high variable signals into unpenalized matrices.

2.4 Modern Deep Learning

The status quo in source separation has shifted away from linear matrix factorization and towards non-linear Deep Learning with neural networks. Architectures such as Bidirectional Long Short-Term Memory (Bi-LSTM) learn the complex temporal and phase reconstructions by processing the data in both directions. [1]

While neural architectures provide superior isolation, they are typically ”black boxes”, require larger datasets, and are computationally expensive. Conversely, the optimization of NMF architectures via decoupled, instrument-specific constraints remains underexplored, and highly interpretable, which is the focus of the subsequent sections!

3 Proposed Approach: Decoupled NMF for Source Separation

To address the previous limitations of uniform regularization in traditional NMF, we propose a supervised, three-stage pipeline. Our framework consists of dictionary pre-training with high-pass filtering, a modified gradient descent optimization phase utilizing decoupled, instrument-specific constraints, and phase-retaining reconstruction using Wiener filtering.

3.1 Data Representation and NMF Formulation The time-domain audio mixture is firstly converted into a time-frequency representation with STFT, resulting in a complex spectrogram:

$$S[F, T] = \begin{bmatrix} A_{1,1}e^{j\phi_{1,1}} & \dots & A_{1n}e^{j\phi_{1,T}} \\ A_{2,1}e^{j\phi_{2,1}} & \dots & A_{2n}e^{j\phi_{2,T}} \\ \vdots & \ddots & \vdots \\ A_{F,1}e^{j\phi_{F,1}} & \dots & A_{F,T}e^{j\phi_{F,T}} \end{bmatrix}$$

Where:

- $A_{F,T} = |S[F, T]| \rightarrow \text{magnitude}$
- $\phi_{F,T} = \angle S[F, T] \rightarrow \text{phase}$

To satisfy the non-negativity constraint required by NMF, we extract the magnitude spectrogram (V), where F is the number of frequency bins and T is the number of time frames. The NMF model approximates our magnitude spectrogram as the linear combination of non-negative dictionaries and their temporal activations:

$$V \approx BW \quad (1)$$

Here, the target sources are defined as drums, bass, vocals, and other. For each source, B corresponds to the spectral basis matrix (dictionary) with a specific rank (K), and W corresponds to the temporal activation matrix, where $B \in \mathbb{R}^{F \times K}$ and $W \in \mathbb{R}^{K \times T}$. Our global dictionary is the concatenation of these individual basis matrices. We minimize the cost function based on KL divergence:

$$D_{KL}(V||BW) = \sum_{i,j} \left(V_{i,j} \log \frac{V_{i,j}}{(BW)_{i,j}} - V_{i,j} + (BW)_{i,j} \right)$$

Minimizing this objective function yields the standard multiplicative update rules. For the Basis Matrix (B):

$$B \leftarrow B \odot \frac{(V/BW)W^T}{\mathbf{1}W^T}$$

For the Activation Matrix (W):

$$W \leftarrow W \odot \frac{B^T(V/BW)}{B^T\mathbf{1}}$$

3.2 Supervised Dictionary Training and Coherence Mitigation

In the first phase, we train the basis matrix for each target source using isolated instrument stems from the MUSDB18 dataset. By isolating the training data, we ensure that, for instance, the drum dictionary strictly models percussive elements, while the vocal dictionary models human speech. We heuristically assigned ranks to prevent overfitting (e.g. $K = 20$ for bass since it is monophonic, and $K = 110$ to capture the massive harmonic diversity of the "other" category).

A critical flaw with supervised NMF is basis coherence, where overlapping frequency subspaces cause energy leakage. To avoid this issue, we enforce orthogonality by applying a high-pass filter to the dictionaries. Frequency bins below 80 Hz in the vocal dictionary and 60 Hz in the "other" dictionary are set to a negligible constant, followed by L1 column re-normalization. [7] This explicitly strips the vocal dictionary of its capacity to reconstruct bass notes.

3.3 Decoupled Regularization and Multiplicative Updates

During the separation phase, the global dictionary (containing drums, bass, vocals, and other) is held strictly constant. The objective is to then optimize W by minimizing the KL divergence between the original audio mixture and our reconstruction.

To prevent penalty collapse, we append targeted penalty functions exclusively to the acoustic characteristics that require them. For example, transient sources such as drums were applied the L_1 sparsity penalty to suppress the noise floor. For harmonic sources like bass and "other", we apply the variance-normalized Temporal Continuity penalty to enforce smooth notes without altering the scale of the activations:

$$c_t(W) = \sum_j \frac{1}{\sigma^2} \sum_t (W_{t,j} - W_{t-1,j})^2 \mid \sigma_j = \sqrt{\frac{1}{T} \sum_{t=1}^T W_{t,j}^2}$$

Conceptually minimizing this reduces the difference between adjacent time frames. Because the derivative of a sum is linear, we calculate the new adaptive learning rate by decomposing the total gradient into its strictly positive and negative components. This yields a modified Multiplicative Update rule for the activation matrices, where the negative regularization gradients are added to the numerator, and the positive regularization gradients are added to the denominator:

$$W_{new} = W_{old} \odot \frac{[\nabla_W C_{base}]^- + [\nabla_W J_{penalty}]^-}{[\nabla_W C_{base}]^+ + [\nabla_W J_{penalty}]^+}$$

The mathematical proof and exact multiplicative updates of our implementation will be detailed in Appendix A.

By tuning the hyperparameters individually (e.g., applying low sparsity and low TC to vocals to preserve jagged syllables, high sparsity to drums to correlate transient natures), we force our algorithm to reject transience in harmonics, and vice versa, improving separation fidelity.

3.4 Phase Recovery via Wiener Filtering

The final raw NMF estimate BW lacks phase information as we did our iterative scheme on a magnitude spectrogram, resulting in robotic audio if reconstructed directly. To resolve this, we utilize the estimate instead to construct a soft mask with the Wiener Filter, representing the proportional energy contribution of each instrument at every time-frequency bin:

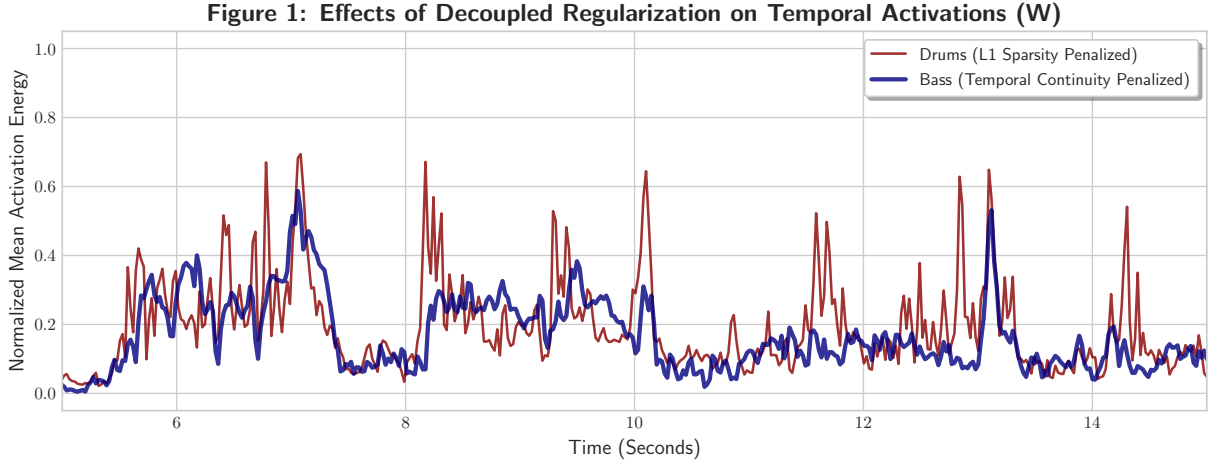
$$\text{Mask} = \frac{\text{Target Estimate}}{\text{Total Combined Estimate}}$$

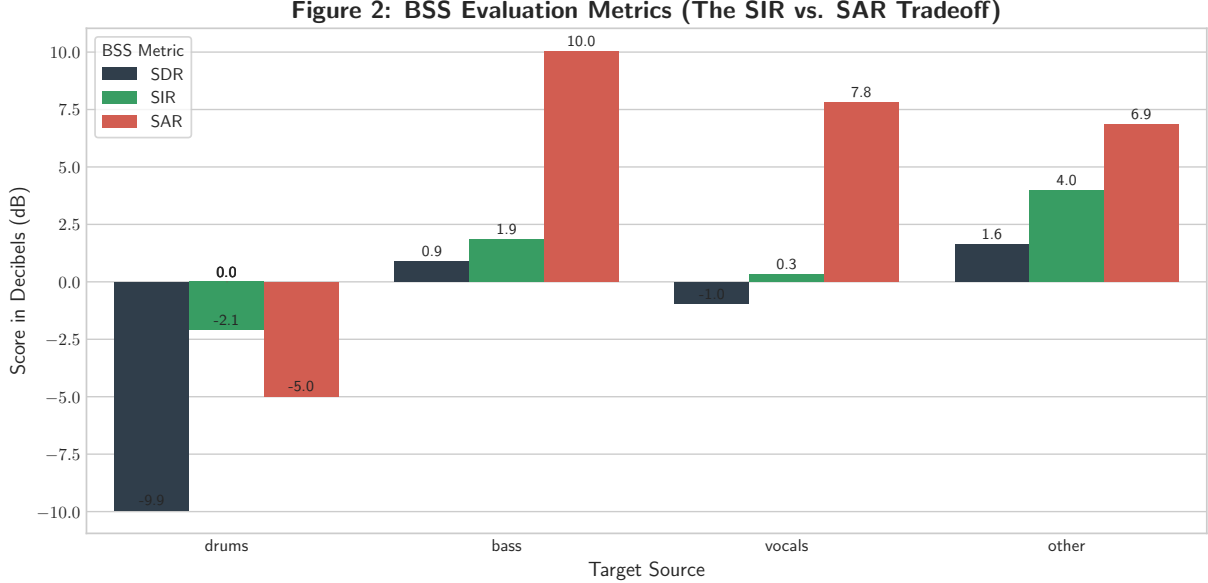
Our isolated stem is the finally the Hadamard product of the fractional mask and the original complex spectrogram, $\text{Mask} \odot X_{\text{mix}}$. This operation extracts the correct target amplitudes while preserving the original phase information. We then finally recover the time-domain signal via the Inverse Short-Time Fourier Transform (ISTFT), completing the pipeline.

4 Results

The source separation pipeline was evaluated using the MUSDB18 dataset. Quantitative performance was measured using the Blind Source Separation (BSS) Evaluation toolkit, calculating the Source-to-Distortion Ratio (SDR), Source-to-Interference Ratio (SIR), and Source-to-Artifact Ratio (SAR) in decibels (dB).

To observe the effects of the decoupled penalty constraints prior to Wiener filtering, the temporal activation matrices (W) were extracted and normalized. The figure below displays the mean activation energy for the drum and bass components over a 10-second sample window. The drum matrix, constrained by an L_1 sparsity penalty of $\lambda_{L1} = 0.05$. The bass matrix was constrained by a temporal continuity penalty of $\lambda_{TC} = 0.85$.





5 Discussion

5.1 Interpretation/Analysis

As illustrated in Figure 1, for a higher L_1 regularization constant, the drum activation energies spike and fall down shortly after, modeling the physical transience of percussive instruments. The high Temporal Continuity penalty models the bass to contain sustained envelopes of the bass, depicting less drastic changes.

The Cost of Sparsity Figure 2 demonstrates that over-penalizing the drums mathematically suppressed its activation, indicated by a drastically low SDR score. Its SIR is higher because it rejects sustained harmonic transitions, but destroys the natural acoustic delay of instruments, reflected by the low SAR score.

SAR values We see that for the majority of the isolated stem performance, the Signal-to-Artifact Ratio score is extremely high. SAR is defined to measure the level of distortion from artificial noise and/or interference. The high performance indicates that reconstructing post soft-masking preserves the original sound information, contrary to reconstructing on the NMF estimate.

The Limits of High-Pass Filtering While we prevented vocals from reconstructing at lower-frequencies, harmonic instruments (guitars, synths, etc.) inherently share overlapping mid-range frequencies, indicated by a small negative SDR value.

5.2 The Limits of the Current Pipeline

The heuristic approach in hard-coding ranks for each stem basis fails to allow room for interpretation in creating a true global representation basis. Additionally, because NMF operates strictly on magnitude, we had to re-inject the original mixture’s phase. When two instruments overlap at the same frequency bin, using the mixed phase introduces interference in the isolated stems.

NMF is additionally a purely linear model, composed of additive combinations of matrices. For non-linear audio features such as dynamic range compression, NMF does not preserve this constraint.

5.3 Implications/Insights

The findings prove that uniform regularization is flawed and assumes each isolated stem is similar in acoustic variance. Sound sources possess contradictory properties, proving that decoupling is necessary for separation, though optimal hyperparameter tuning remains a subject for future work. And while neural networks achieve vastly superior performance for SDR, this approach offers interpretability, providing mathematical intuition to acoustic traits.

6 Conclusion

In this paper, for separating different music stems from an audio mixture, we proposed a novel three-stage implementation to improve the traditional non-negative matrix factorization algorithm. We first heuristically define the ranks of each isolated stem basis, proportionally representing the acoustic variance of each instrument. We then apply a high-pass filter to the basis and normalize it to represent the attainable frequency of vocals. Furthermore, we developed an adaptive gradient descent iterative method, adding a sparse and temporal cost term. Lastly, with our NMF estimation, we apply a Wiener filter to the original spectrogram matrix to isolate our music stems. Our experiments convey that different regularization constants capture more flexibility to audio variations.

While the decoupled regularization framework successfully demonstrates the necessity of instrument-specific constraints, several avenues for future research could address the current limitations of the pipeline:

- **Adaptive Rank Selection:** The current methodology relies on heuristically hard-coded matrix ranks (K) for each source. Future iterations should explore data-driven rank determination, such as Automatic Relevance Determination (ARD) to dynamically allocate basis vectors based on the acoustic complexity of the input mixture.
- **Convolutional NMF (CNMF):** Convolutional non-negative matrix factorization incorporates convolutional operations to model temporally or spatially shifted, repeating patterns in data. This removes the need for the temporal cost function, and inherently captures more underlying patterns in its implementation.

- **Automated Hyperparameter Tuning:** The current regularization weights (λ_{L1} and λ_{TC}) were manually tuned to expose specific mathematical tradeoffs. Implementing grid search algorithms over a larger validation dataset would allow for the discovery of mathematically optimal constants that maximize BSS evaluation metrics without inducing penalty collapse.
- **Hybridizing NMF with Deep Learning:** To overcome the purely linear constraints of NMF and address non-linear mixing effects like dynamic range compression, we can combine NMF with deep learning. By using the decoupled NMF activations as interpretable prior inputs for a Convolutional Neural Network (CNN), one could theoretically combine the mathematical transparency of NMF with the superior non-linear separation capabilities of modern neural architectures.

A Mathematical Derivations

A.1 Multiply Update from KL Divergence

In standard Non-Negative Matrix Factorization (NMF) utilizing the Kullback-Leibler (KL) divergence, the gradient of the cost function with respect to the activation matrix W (where $V \approx BW$) is derived as follows.

First, expand the KL divergence cost function to isolate the terms containing W :

$$\begin{aligned} D_{KL}(V||BW) &= \sum_{i,j} \left(V_{ij} \log \frac{V_{ij}}{(BW)_{ij}} - V_{ij} + (BW)_{ij} \right) \\ &= \sum_{i,j} \left(V_{ij} \log V_{ij} - V_{ij} \log \left(\sum_k B_{ik} W_{kj} \right) - V_{ij} + \sum_k B_{ik} W_{kj} \right) \end{aligned}$$

Take the partial derivative with respect to a single element W_{aj} :

$$\frac{\partial D_{KL}}{\partial W_{aj}} = \sum_i \left(-V_{ij} \frac{1}{\sum_k B_{ik} W_{kj}} \frac{\partial}{\partial W_{aj}} \left(\sum_k B_{ik} W_{kj} \right) + \frac{\partial}{\partial W_{aj}} \left(\sum_k B_{ik} W_{kj} \right) \right)$$

Since $\frac{\partial}{\partial W_{aj}} (\sum_k B_{ik} W_{kj}) = B_{ia}$, substituting this in yields:

$$\begin{aligned} \frac{\partial D_{KL}}{\partial W_{aj}} &= \sum_i \left(-V_{ij} \frac{B_{ia}}{\sum_k B_{ik} W_{kj}} + B_{ia} \right) \\ &= \sum_i B_{ia} - \sum_i V_{ij} \frac{B_{ia}}{(BW)_{ij}} \end{aligned}$$

Converting this element-wise derivative back into its full matrix form gives the total gradient:

$$\nabla_W D_{KL} = B^\top \mathbf{1} - B^\top \left(\frac{V}{BW} \right)$$

Standard gradient descent updates the matrix using an additive step with a learning rate η :

$$\begin{aligned} W^{t+1} &\leftarrow W^t - \eta \odot \nabla_W D_{KL} \\ &\leftarrow W^t - \eta \odot \left(B^\top \mathbf{1} - B^\top \left(\frac{V}{BW} \right) \right) \end{aligned}$$

Referencing Lee and Seung's adaptive learning rate methodology to enforce non-negativity, we define $\eta = \frac{W^t}{B^\top \mathbf{1}}$:

$$\begin{aligned} W^{t+1} &\leftarrow W^t - \frac{W^t}{B^\top \mathbf{1}} \odot \left(B^\top \mathbf{1} - B^\top \left(\frac{V}{BW} \right) \right) \\ W^{t+1} &\leftarrow W^t - W^t \odot \frac{B^\top \mathbf{1}}{B^\top \mathbf{1}} + W^t \odot \frac{B^\top \left(\frac{V}{BW} \right)}{B^\top \mathbf{1}} \end{aligned}$$

The subtraction perfectly cancels out the current state ($W^t - W^t$), leaving the final multiplicative update rule:

$$W \leftarrow W \odot \frac{B^\top \left(\frac{V}{BW} \right)}{B^\top \mathbf{1}}$$

The same line of reason follows for the multiplicative update for B .

A.2 Multiplicative Update for Separation Stage

During the separation stage, when you add penalties to NMF, the total cost function becomes: $C_{total} = D_{KL}(V||BW) + J_{L_1}(W) + J_{TC}(W)$. By construction, derivatives are linear, so we take the derivative of each penalty separately and add them to the base KL gradient.

1. The L_1 Sparsity Penalty

The cost function for sparsity is the sum of all elements multiplied by a weight λ_{L_1} :

$$J_{L_1} = \lambda_{L_1} \sum_{k,t} W_{k,t}$$

The derivative with respect to any single element $W_{k,t}$ is just the λ_{L_1} .

- **Positive Component:** λ_{L_1}

- **Negative Component:** 0

2. The Temporal Continuity (TC) Penalty

The cost function penalizes the squared differences between adjacent time frames. Including the regularization weight λ_{TC} , the objective function is:

$$J_{TC} = \lambda_{TC} \sum_k \frac{\sum_t (W_{k,t} - W_{k,t-1})^2}{\frac{1}{T} \sum_t W_{k,t}^2}$$

To simplify the application of the quotient rule, let the numerator be $N_k = \sum_t (W_{k,t} - W_{k,t-1})^2$ and the variance-normalizing denominator be $D_k = \frac{1}{T} \sum_t W_{k,t}^2$.

When taking the partial derivative with respect to a single element $W_{k,t}$, we evaluate the numerator and denominator independently:

Derivative of the Numerator (N_k):

Using the chain rule on the connection between the preceding and succeeding frames:

$$\begin{aligned} \frac{\partial N_k}{\partial W_{k,t}} &= 2(W_{k,t} - W_{k,t-1}) + 2(W_{k,t} - W_{k,t+1}) \\ &= 4W_{k,t} - 2W_{k,t-1} - 2W_{k,t+1} \end{aligned}$$

Derivative of the Denominator (D_k):

$$\frac{\partial D_k}{\partial W_{k,t}} = \frac{2}{T} W_{k,t}$$

Applying the quotient rule $\frac{N'_k D_k - N_k D'_k}{D_k^2}$ to the full cost function yields:

$$\nabla_{W_{k,t}} J_{TC} = \lambda_{TC} \left[\frac{1}{D_k} (4W_{k,t} - 2W_{k,t-1} - 2W_{k,t+1}) - \frac{N_k}{D_k^2} \left(\frac{2}{T} W_{k,t} \right) \right]$$

To map this to Lee and Seung's multiplicative updates, we isolate the strictly positive and strictly negative additive components of the gradient:

- **Positive Component:** $[\nabla_{W_{k,t}} J_{TC}]^+ = \lambda_{TC} \left(\frac{4}{D_k} W_{k,t} \right)$
- **Negative Component:** $[\nabla_{W_{k,t}} J_{TC}]^- = \lambda_{TC} \left(\frac{2}{D_k} (W_{k,t-1} + W_{k,t+1}) + \frac{2N_k}{TD_k^2} W_{k,t} \right)$

3. The Final Modified Update Rule:

By substituting these decomposed regularization components into the base Lee and Seung adaptive learning rate substitution derived in the previous section, we arrive at our final, stabilized update rule:

$$W \leftarrow W \odot \frac{B^\top \left(\frac{V}{BW} \right) + \lambda_{TC} \left(\frac{2}{D_k} (W_{\leftarrow} + W_{\rightarrow}) + \frac{2N_k}{TD_k^2} W \right)}{B^\top \mathbf{1} + \lambda_{L_1} + \lambda_{TC} \left(\frac{4}{D_k} W \right)}$$

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